Multi-waveguide indoor positioning for PASS

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***Abstract*—The multi-waveguide-based uplink indoor positioning approaches for pinching-antenna (PA) systems (PASS) are first proposed in this article. Two possible scenarios, multi-waveguide single-PA (MWSP) and multi-waveguide multi-PA (MWMP) scenarios, are considered. For the MWSP scenario, the received signal strength indication-based ranging method and the MWSP-based least square (LS) positioning algorithm were developed. To gain deeper insights, a comprehensive error analysis of the LS positioning algorithm is conducted. Subsequently, for the MWMP scenario, the close-form expression of the superposed signal is derived. According to the signal power, the MWMP-based grid search algorithm is proposed and the estimation error of proposed algorithm is analyzed. Numerical results validate the correctness of our analysis, which show that: i) For the MWSP scenario, a smaller geometric dilution of precision (GDoP) leads to a lower average positioning error. Furthermore, even when the GDoP is large, the regions where the distances to PAs are nearly equal achieve the best accuracy. ii) For the MWMP scenario, non-parallel waveguide deployment improves positioning accuracy, although errors increase with the number of PAs.**

Index Terms*—Grid search, LS, MWSP, MWMP, RSSI, PASS.

I. Introduction

As a promising technique for overcoming blockage and restoring line of sight (LoS) connections in dynamic wireless environments, pinching-antenna (PA) systems (PASS) have been proposed [1]. By pinching the PA at an arbitrary position along the low-loss waveguide, a radiating element can be formed at that point [2]. Recently, many researchers have explored the performance analysis and optimization of PASS. Wang et al. investigated a non-orthogonal multiple access assisted downlink PA system, which can activate multiple PAs at predefined locations on a dielectric waveguide to support users [3]. Ouyang et al. presented a closed-form characterization of the array gain limit in PASS and explored its variation with respect to antenna spacing. Simulation results show that appropriate configuration of antenna quantity and inter-element spacing allows PASS to deliver superior array gain compared with traditional antenna deployments [4]. In addition, Zhou et al. proposed a gradient-based meta-learning joint optimization algorithm, in which the joint optimization of beamforming design and antenna deployment was performed to maximize the weighted sum rate [5].

Despite the promising performance of PASS in enhancing communication efficiency, its potential for indoor positioning remains largely unexplored. Existing indoor positioning methods generally utilize wireless fidelity, ultra wide band (UWB) and fixed antenna arrays [6], [7]. Recently, many researchers have proposed the reconfigurable intelligence surface (RIS)-assisted indoor positioning systems. Pu et al. built a virtual LoS link through RIS, which improved the positioning performance in non line of sight environments [8]. Ma et al. applied RIS to indoor positioning with UWB technique and derived the Cramér–Rao limit for the proposed localization approach [9]. Importantly, zhang et al. proposed a PASS-based indoor positioning approach and developed a positioning algorithm based on a single waveguide with multi-PA, where the proposed method is efficient and particularly applicable to PASS [10].

Based on the previous analysis, it is evident that the majority of studies have concentrated on PASS for communications, whereas the research on indoor positioning remains relatively unexplored. Existing studies on PASS generally assume that user locations are known when optimizing PA positions, leaving the fundamental challenge of acquiring positioning information unaddressed. In particular, single-waveguide-based indoor positioning methods suffer from the limitation that only one PA can be activated in each time slot, resulting in poor real-time positioning performance. Moreover, indoor positioning approaches for multi-waveguide PASS have not yet been explored. Even in current uplink multi-waveguide schemes, each waveguide is typically restricted to a single PA, and the critical issue of uplink signal superposition under simultaneous multi-PA activation remains unresolved.

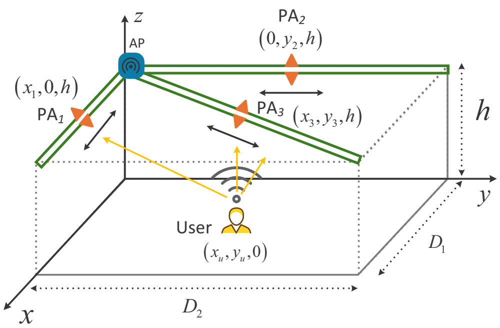

Fig. 1. MWSP system model.

To overcome these limitations, this paper investigates the comprehensive uplink positioning techniques for multi-waveguide PASS and develops dedicated positioning algorithms for both multi-waveguide single-PA (MWSP) and multi-waveguide multi-PA (MWMP) scenarios, thereby enhancing the applicability of PASS in indoor positioning.

II. MWSP System Model And Positioning Approach

This section focuses on the system model and the proposed positioning approach in the MWSP scenario. We first present a detailed description of the system architecture. Then, we propose the positioning approach.

A. MWSP System Model

1) Antenna Model: As depicted in Fig. 1, the room is characterized by dimensions $x \in [0, D_1]$, $y \in [0, D_2]$, and $z \in [0, h]$, with an access point (AP) located at the ceiling corner. Three waveguides are connected to the AP, extending along the ceiling edges of x-axis, the ceiling edges of y-axis, and the diagonal of the ceiling, respectively. The position of three waveguides are defined as $W_1 \in ((0, D_1), 0, h)$, $W_2 \in (0, (0, D_2), h)$ and $W_3 \in ((0, D_1), (0, D_2), h)$, respectively. Each waveguide hosts one PA, with the position of the k-th PA denoted as $L_k = (x_k, y_k, h)$ for $k = 1, 2, 3$, which is considered to be known. The position of the user is defined as $(x_u, y_u, 0)$, which is the coordinate to be solved.

2) Positioning Channel and Signal Model: To simplify the theoretical investigation of performance limits and asymptotic behavior, the channel is modeled based on the LoS assumption. According to the free space propagation theorem, the large-scale channel can be written as:

$$h_{ls}(d_{uk}) = \frac{c}{4\pi f_c d_{uk}}, \tag{1}$$

where c represents the speed of light and f_c is the carrier frequency. d_{uk} represents the distance between the user and PA of the k-th waveguide, which can be written as:

$$d_{uk} = \sqrt{(x_k - x_u) + (y_k - y_u)^2 + h^2}. \tag{2}$$

For the small-scale fading component, the channel coefficients are normalized, with analysis focused solely on the phase shift resulting from free-space transmission. The small-scale channel is modelled as:

$$h_{ss}(d_{uk}) = \exp\left(-j\frac{2\pi}{\lambda}d_{uk}\right), \tag{3}$$

where λ denotes the wave length.

When the signal propagates in the waveguide, we use the complex propagation constant γ to characterize the phase shift and amplitude attenuation per unit length, which is given by [11]:

$$\gamma = \alpha + j\beta, \tag{4}$$

where α denotes the attenuation constant accounting for dielectric and conductor losses, and β is the phase constant determined by the waveguide geometry and material properties. Specifically, α and β can be respectively written as:

$$\alpha = \frac{\pi\sqrt{\varepsilon_r}\tan\delta}{\lambda}, \tag{5}$$

and

$$\beta = \frac{2\pi\sqrt{\varepsilon_r}}{\lambda}, \tag{6}$$

where ε_r and $\tan\delta$ denote relative permittivity and loss angle tangent, respectively.

Thus, the channel model of the k-th waveguide can be written as:

$$\begin{aligned} h_w(x_k, y_k) = \exp&\left(-\frac{\pi\sqrt{\varepsilon_r}\tan\delta}{\lambda}\left(\sqrt{x_k^2 + y_k^2}\right)\right) \\ &\times\exp\left(-j\frac{2\pi\sqrt{\varepsilon_r}}{\lambda}\left(\sqrt{x_k^2 + y_k^2}\right)\right). \end{aligned} \tag{7}$$

The signal undergoes free-space propagation, couples into the waveguide through the PA, and is subsequently guided to the AP. The signal received from the k-th waveguide is given by:

$$\begin{aligned} r_k = \frac{\sqrt{P_s}c}{4\pi f_c d_{uk}}&\exp\left(-\frac{\pi\sqrt{\varepsilon_r}\tan\delta}{\lambda}\left(\sqrt{x_k^2 + y_k^2}\right)\right) \\ \times\exp&\left(-j\frac{2\pi}{\lambda}\left(d_{uk} + \sqrt{\varepsilon_r}\left(\sqrt{x_k^2 + y_k^2}\right)\right)\right)s + \eta_k, \end{aligned} \tag{8}$$

where P_s represents the transmission power of the signal, and η_k is the additive white Gaussian noise (AWGN) with mean zero and variance σ^2. s denotes the transmitted signal.

B. Positioning Approach and Error Analysis

For the k-th waveguide, the received power of AP can be written as:

$$P_{rk} = P_s\left(\frac{ce^{-\alpha\sqrt{x_k^2+y_k^2}}}{4\pi f_c d_{uk}}\right)^2 + \xi_k, \tag{9}$$

where $\xi_k \sim \mathcal{N}(0, \sigma^2)$ denotes the power of AWGN.

The RSSI is then employed to estimate the distance between each PA and the user. The noisy estimated distance from the

i-th PA to the user is expressed as:

$$\hat{d}_{uk} = \frac{ce^{-\alpha\sqrt{x_k^2+y_k^2}}}{\sqrt{\frac{P_{rk}}{P_s}}4\pi f_c}. \tag{10}$$

We combine (2) and (10), the corresponding distance measurement equation is then derived as:

$$(x_k - x_u)^2 + (y_k - y_u)^2 + h^2 = \frac{P_s}{P_{rk}}\left(\frac{ce^{-\alpha\sqrt{x_k^2+y_k^2}}}{4\pi f_c}\right)^2. \tag{11}$$

According to (11), the distance equation in MWSP scenario can be written as:

$$(x_1 - x_u)^2 + (y_1 - y_u)^2 + h^2 = \hat{d}_{u1}^2, \tag{12a}$$
$$(x_2 - x_u)^2 + (y_2 - y_u)^2 + h^2 = \hat{d}_{u2}^2, \tag{12b}$$
$$(x_3 - x_u)^2 + (y_3 - y_u)^2 + h^2 = \hat{d}_{u3}^2. \tag{12c}$$

We subtract the first equation (12a) from the subsequent ones (12b) and (12c), and then (12) is linearized for further analysis, which is given by:

$$\begin{cases} 2x_u(x_2 - x_1) + 2y_u(y_2 - y_1) = e_1, \\ 2x_u(x_3 - x_1) + 2y_u(y_3 - y_1) = e_2, \end{cases} \tag{13}$$

where

$$e_1 = \hat{d}_{u1}^2 - \hat{d}_{u2}^2 + \left(x_2^2 - x_1^2\right) + \left(y_2^2 - y_1^2\right), \tag{14}$$

and

$$e_2 = \hat{d}_{u1}^2 - \hat{d}_{u3}^2 + \left(x_3^2 - x_1^2\right) + \left(y_3^2 - y_1^2\right). \tag{15}$$

From a matrix perspective, (13) can be written as:

$$\mathbf{AX} = \mathbf{b}, \tag{16}$$

where

$$\mathbf{A} = \left[\begin{array}{cc} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - x_1) \end{array}\right], \tag{17}$$

$$\mathbf{X} = \left[\begin{array}{cc} x_u & y_u \end{array}\right]^T, \tag{18}$$

$$\mathbf{b} = \left[\begin{array}{cc} e_1 & e_2 \end{array}\right]^T. \tag{19}$$

The position estimation $\mathbf{X}$ is given by:

$$\mathbf{X} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T\mathbf{b}. \tag{20}$$

We then perform a detailed error analysis of the proposed positioning algorithms. The coordinates of the user are calculated by (20), and since $\mathbf{A}$ is a square matrix with a non-zero determinant, (20) can be rewritten as:

$$\begin{aligned} \mathbf{X} &= \mathbf{A}^{-1}\mathbf{b} = \mathbf{A}^{-1}(\mathbf{AX}_t + \mathbf{g}) \\ &= \mathbf{X}_t + \mathbf{A}^{-1}\mathbf{g}, \end{aligned} \tag{21}$$

where $\mathbf{X}_t$ represents the error-free solution vector, and $\mathbf{g}$ denotes error vector.

Therefore, the positioning errors are caused by the error amplification factor $\mathbf{A}^{-1}$ and the error in the observation $\mathbf{b}$.

i) Errors caused by matrix $\mathbf{A}^{-1}$: The inverse of matrix $\mathbf{A}$ can be written as:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\mathbf{A}^*, \tag{22}$$

where $\det(\mathbf{A})$ and $\mathbf{A}^*$ denote the determinant and adjoint matrix of $\mathbf{A}$, respectively.

Thus, the matrix $\mathbf{A}^{-1}$ is inversely proportional to the determinant. According to the expression in (17), the determinant of matrix $\mathbf{A}$ can be written as:

$$\det(\mathbf{A}) = 4\left[(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right]. \tag{23}$$

We define the triangle enclosed by the spatial coordinates (x_1, y_1), (x_2, y_2) and (x_3, y_3) as Δ_p, and we define the area of Δ_p as S_{Δ_p}. According to the definition of the cross product modulus, we can derive that the determinant of $\mathbf{A}$ is eight times the area enclosed by three PAs, i.e., $\det(\mathbf{A}) = 8S_{\Delta_p}$.

Remark 1. *The larger the area enclosed by PAs, the smaller the error amplification matrix $\mathbf{A}^{-1}$, thereby suppressing error amplification and enabling stable, accurate positioning.*

ii) Errors caused by observation vector b: From the definitions in (14), the observation error of e_1 can be written as:

$$\begin{aligned} \Delta e_1 &= \left|\left(\hat{d}_{u1}^2 - d_{u1}^2\right) - \left(\hat{d}_{u2}^2 - d_{u2}^2\right)\right| \\ &= \left|\left((d_{u1} + \delta_1)^2 - d_{u1}^2\right) - \left((d_{u2} + \delta_2)^2 - d_{u2}^2\right)\right| \\ &= \left|\left(2d_{u1}\delta_1 + \delta_1^2\right) - \left(2d_{u2}\delta_2 + \delta_2^2\right)\right| \\ &\approx \left|2(d_{u1}\delta_1 - d_{u2}\delta_2)\right|, \end{aligned} \tag{24}$$

where δ_1, δ_2 represent the distance measurement error, which are much smaller than the distance.

Similarly, the observation error of e_2 can be written as:

$$\Delta e_2 \approx \left|2(d_{u1}\delta_1 - d_{u3}\delta_3)\right|, \tag{25}$$

where δ_3 denotes the distance measurement error of d_{u3}.

Thus, the error vector $\mathbf{g}$ can be approximated as:

$$\mathbf{g} = 2\left[\begin{array}{cc} (d_{u1}\delta_1 - d_{u2}\delta_2) & (d_{u1}\delta_1 - d_{u3}\delta_3) \end{array}\right]^T. \tag{26}$$

Remark 2. *According to remark 1 and the error vector in* (26)*, the positioning accuracy is determined by the area enclosed by the coordinates of three PAs and the distance differences between the user and three PAs.*

III. MWMP System Model And Positioning Approach

This section examines the system model and positioning approach when multiple PAs are activated simultaneously in the same waveguide.

1) Antenna Model: In the MWMP scenario, the spatial dimensions of the room and the waveguide distribution are identical to that of the MWSP scenario. As depicted in Fig. 2, the PA array with N elements is located on each waveguide to serve the randomly distributed user, where the position of the user is the same as the MWSP scenario. Without loss of generality, it is assumed that N is an odd integer, then the

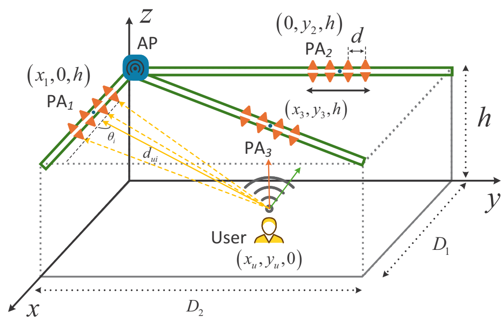

Fig. 2. MWMP system model.

center of the array is the middle PA. The location of PA array center of the i-th waveguide is defined as (x_i, y_i, h), and the distance d between the PA elements is identical.

2) Positioning Channel and Signal Model: Similar to (1), the large-scale channel between the user and the i-th waveguide can be modeled as:

$$h_{ls}(d_{ui}) = \frac{c}{4\pi f_c d_{ui}}, \tag{27}$$

where d_{ui} represents the distance between the user and the array center of the i-th waveguide, which can be written as:

$$d_{ui} = \sqrt{(x_i - x_u) + (y_i - y_u)^2 + h^2}. \tag{28}$$

The small-scale channel can be written as:

$$h_{ss}(d_{in}) = \exp\left(-j\frac{2\pi}{\lambda} d_{in}\right), \tag{29}$$

where d_{in} denotes the distance from the user to the n-th PA in the array of the i-th waveguide, which is given by:

$$d_{in} = \sqrt{d_{ui}^2 + n^2 d^2 - 2 d_{ui} n d \cos\theta_i}, \tag{30}$$

where $n = -\frac{N-1}{2}, ..., 0, ..., \frac{N-1}{2}$. θ_i denotes the angle between signal and PA array center of the i-th waveguide, with $\theta_i \in (0, \pi)$.

The channel model of the i-th waveguide is given by:

$$\begin{aligned} h_w(x_i, y_i) = &\exp\left(-\frac{\pi\sqrt{\varepsilon_r}\tan\delta}{\lambda}\left(\sqrt{x_i^2 + y_i^2} + nd\right)\right) \\ &\times \exp\left(-j\frac{2\pi\sqrt{\varepsilon_r}}{\lambda}\left(\sqrt{x_i^2 + y_i^2} + nd\right)\right). \end{aligned} \tag{31}$$

The signal received from the i-th waveguide in AP can be expressed as:

$$r_i = \sqrt{P_s}\frac{c}{4\pi f_c d_{ui}} \sum_{n=-(N-1)/2}^{(N-1)/2} E_n^i s + \eta_i, \tag{32}$$

where

$$\begin{aligned} E_n^i = &\exp\left(-\frac{\pi\sqrt{\varepsilon_r}\tan\delta}{\lambda}\left(\sqrt{x_i^2 + y_i^2} + nd\right)\right) \\ &\times \exp\left(-j\frac{2\pi}{\lambda}\left(d_{in} + \sqrt{\varepsilon_r}\left(\sqrt{x_i^2 + y_i^2} + nd\right)\right)\right). \end{aligned} \tag{33}$$

A. Positioning Approach and Error Analysis

From (32) and (33), we can observe that the power of the received signal is related the user's position. In order to simplify the analysis and obtain the closed-form expression of the superimposed received signal, we define $\sqrt{\varepsilon_r} d = \lambda$. Then, $d = \lambda/\sqrt{\varepsilon_r}$, and the distance d satisfies $d \ll 1$. Therefore, the waveguide loss from each PA to the AP can be approximated as the loss from the array center to AP. Then, expanding the distance d_{in} in (30) by using a first-order Taylor series, the linearized distance is derived as follows:

$$d_{in} = d_{ui} - nd\cos\theta_i. \tag{34}$$

Therefore, the superimposed received signal in (32) can be rewritten as:

$$\begin{aligned} r_i = &\frac{\sqrt{P_s} c e^{-\alpha\sqrt{x_i^2 + y_i^2}}}{4\pi f_c d_{ui}} \\ &\times \exp\left(-j\frac{2\pi}{\lambda}\left(d_{ui} + \sqrt{\varepsilon_r}\sqrt{x_i^2 + y_i^2}\right)\right) \\ &\times \sum_{n=-(N-1)/2}^{(N-1)/2} \exp\left(j\frac{2\pi}{\lambda} nd\cos\theta_i\right) s + \eta_i. \end{aligned} \tag{35}$$

The cosine values of angles between the signal and three waveguides can be respectively expressed as:

$$\cos\theta_1 = \frac{x_u - x_1}{d_{u1}}, \tag{36}$$

$$\cos\theta_2 = \frac{y_u - y_2}{d_{u2}}, \tag{37}$$

and

$$\cos\theta_3 = \frac{D_1(x_u - x_3) + D_1(y_u - y_3)}{d_{u2}\sqrt{D_1^2 + D_2^2}}. \tag{38}$$

Lemma 1. *For the waveguide distributed on the ceiling, the closed-form expression of the superimposed received signal can be written as:*

$$\begin{aligned} r_i = &\frac{\sqrt{P_s} c e^{-\alpha\sqrt{x_i^2 + y_i^2}}}{4\pi f_c d_{ui}} \frac{\sin\left(\frac{N\psi_i}{2}\right)}{\sin\left(\frac{\psi_i}{2}\right)} \\ &\times \exp\left(-j\frac{2\pi}{\lambda}\left(d_{ui} + \sqrt{\varepsilon_r}\sqrt{x_i^2 + y_i^2}\right)\right) s + \eta_i, \end{aligned} \tag{39}$$

where

$$\psi_i = \frac{2\pi d\cos\theta_i}{\lambda}.$$

Proof. We focus on the summation term in (35). Since $\psi_i =$

$\frac{2\pi d\cos\theta_i}{\lambda}$, the summation term can be written as:

$$\begin{aligned}&\sum_{n=-(N-1)/2}^{(N-1)/2}\exp\left(jn\psi_i\right)=\frac{e^{-j\frac{N-1}{2}\psi_i}-e^{j\frac{N+1}{2}\psi_i}}{1-e^{j\psi_i}}\\&=\frac{e^{j\frac{\psi_i}{2}}\left(e^{-j\frac{N\psi_i}{2}}-e^{j\frac{N\psi_i}{2}}\right)}{e^{j\frac{\psi_i}{2}}\left(e^{-j\frac{\psi_i}{2}}-e^{j\frac{\psi_i}{2}}\right)}=\frac{\sin\left(\frac{N\psi_i}{2}\right)}{\sin\left(\frac{\psi_i}{2}\right)}.\end{aligned} \tag{40}$$

By substituting (40) into (35), we can obtain the closed-form expression in (39). The proof is complete. □

The power of the received signal can be derived as follow:

$$P_{thi}=P_s\left(\frac{ce^{-\alpha\sqrt{x_i^2+y_i^2}}}{4\pi f_c d_{ui}}\frac{\sin\left(\frac{N\psi_i}{2}\right)}{\sin\left(\frac{\psi_i}{2}\right)}\right)^2+\zeta_i, \tag{41}$$

where $\zeta_i \sim \mathcal{N}\left(0,\sigma^2\right)$ denotes the power of AWGN.

Assuming that the measured power is P_{ri}, the superimposed power-based positioning equation can be given by:

$$P_{ri}\left(x_u,y_u\right)=P_s\left(\frac{ce^{-\alpha\sqrt{x_i^2+y_i^2}}}{4\pi f_c d_{ui}}\frac{\sin\left(\frac{N\psi_i}{2}\right)}{\sin\left(\frac{\psi_i}{2}\right)}\right)^2+\zeta_i. \tag{42}$$

However, since $\cos\theta_i$ is symmetrical in $\theta_i \in (0,\pi)$, $\left(\frac{\sin\left(\frac{N\psi_i}{2}\right)}{\sin\left(\frac{\psi_i}{2}\right)}\right)^2$ is a symmetric function. Therefore, the received power is also symmetrical.

Remark 3. *When multiple parallel waveguides are distributed on the ceiling, the positioning performance is poor because parallel waveguides share identical symmetry forms, differing only in symmetry points, making the symmetry difficult to eliminate.*

Algorithm 1 MWMP-based Grid Search Algorithm

Input: Coordinates of PA array centers, transmitted power, and measured received power of each waveguide.

Output: Estimated user coordinate $(\hat{x}_u, \hat{y}_u, 0)$.

1: Define error function $e_r\left(x_u,y_u\right)=\sum_{i=1}^{3}\left|P_{ri}-P_{thi}\right|$.

2: Divide the room into coarse grid points and compute received power P_{ri} for each grid point by (41).

3: Evaluate the error function $e_r\left(x_u,y_u\right)$ at each grid point to find the minimal-error point.

4: Apply iterative refinement by subdividing the area around the grid point with minimal initial error to form a finer search grid, where $x_u \in [0, D_1]$ and $y_u \in [0, D_2]$.

5: Repeat the evaluation of the error function over the refined grid, progressively reducing grid size to achieve the desired localization resolution.

6: **return**: The coordinate of the grid point $\hat{x}_u$, $\hat{y}_u$.

Thus, multiple non-parallel waveguides are required to break the symmetry and enable precise positioning for user. When the waveguides are not distributed in parallel, the superposition power exhibits different symmetry, which inspired us to design a multi-waveguide joint positioning algorithm. Algorithm 1 presents the specific steps of the MWMP-based grid search algorithm.

We then analyzed the error sources of the proposed positioning approach. First, the waveguide attenuation is approximated by that from the PA array center to the AP. Since the waveguide loss is much smaller than the free-space path loss and the inter-PA distance satisfies $d \ll 1$, its effect on signal amplitude is negligible. Second, the first-order Taylor approximation in (34) simplifies the analysis but introduces phase errors due to neglected higher-order terms, leading to discrepancies in the theoretical superimposed power. Finally, although Algorithm 1 employs three waveguides, certain spatial configurations may still yield symmetric power distributions, causing localization ambiguity. As the number of PAs increases, the intensified sidelobes of the theoretical power further aggravate these errors and reduce positioning accuracy.

Remark 4. *Due to the power side lobes, the positioning performance of the proposed algorithm decreases as the number of PAs increases.*

Remark 5. *Since non-parallel waveguides have different symmetry characteristics, combining non-parallel waveguides for positioning can improve accuracy.*

IV. Simulation Results

Here, simulations are performed to evaluate the performance of MWSP-based LS algorithm and MWMP-based grid search algorithm. The room dimensions are defined as $x \in [0,6]$, $y \in [0,10]$, and $z \in [0,3]$. A carrier frequency of 2.4 GHz is considered, while the user's transmit power is fixed at 0.1 W. The dielectric properties are characterized by a relative permittivity of $\varepsilon_r = 2.08$ and a loss tangent of $\tan\delta = 0.0004$ [11].

A. Positioning performance in MWSP scenario

In the MWSP scenario, each waveguide is equipped with one PA, and three PAs are activated simultaneously. We conduct simulations to evaluate the impact of PA distribution on positioning accuracy.

The impact of PAs distribution on positioning accuracy: Fig. 3 demonstrates the impact of the spatial deployment of PAs on positioning errors. Magenta stars represent the coordinates of PAs. When the PAs are sparsely distributed within the room, the positioning performance within the region enclosed by PAs is superior. Moreover, the larger the enclosed area, the smaller the positioning error, which verifies our **Remark 1**. In contrast, when the PAs are more densely distributed within the room, the region with smaller positioning error is not inside the region enclosed by PAs, but in the region with nearly equal distance to three PAs, which verifies our **Remark 2**.

B. Positioning performance in MWMP scenarios

In the MWMP scenario, N PAs are located on the waveguide along x-axis, y-axis and diagonal, respectively. The PAs are activated simultaneously. We conduct simulations to verify

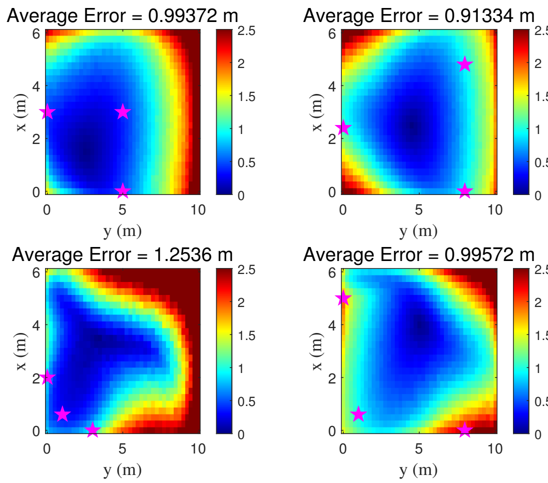

Fig. 3. Positioning error distribution for various PA deployment, with σ^2 = -40 dBm.

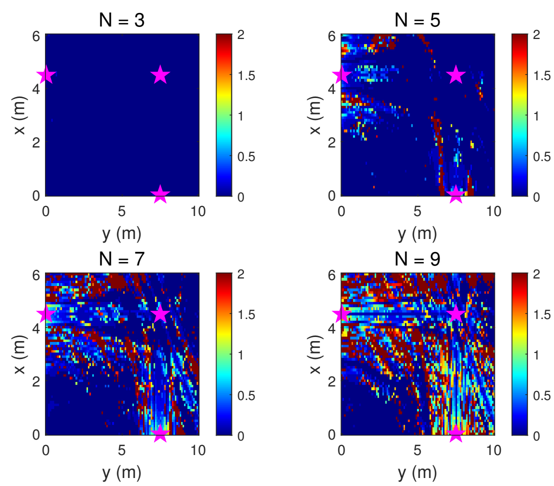

Fig. 4. Positioning errors under different numbers of PAs, and the noise power is set to σ^2 = -80 dBm.

the effectiveness of the proposed MWMP-based grid search algorithm.

1) Impact of the number of PAs on positioning accuracy: Fig. 4 illustrates the impact of the number of PAs on positioning performance. As the number of PAs on each waveguide increases, the performance gradually degrades due to the emergence of stronger sidelobes in the theoretical power function, which introduce additional symmetry points and cause positioning ambiguity. The simulation results validate our **Remark 3** and **Remark 4**. Notably, even when the number of PAs reaches nine, the positioning error remains small in the strip-shaped region connecting the array center and the lower-left area. This is because these regions are constrained by multiple geometric dimensions, resulting in significantly higher positioning accuracy, consistent with our previous findings.

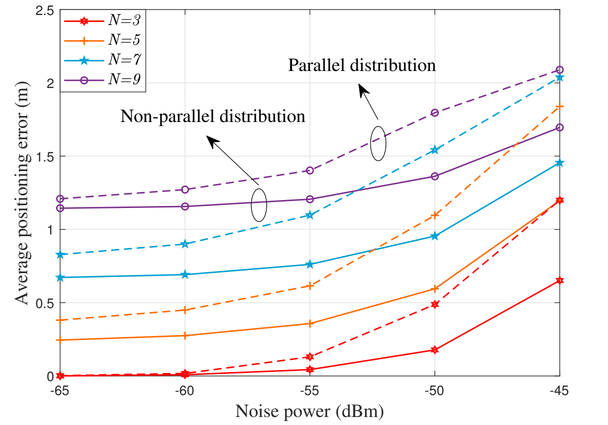

Fig. 5. Average positioning errors of parallel and non-parallel waveguide distributions.

2) Comparison of positioning performance between parallel and non-parallel waveguide distribution: As depicted in Fig. 5, the non-parallel waveguide distribution consistently outperforms the parallel distribution, and the gap widens when the noise power is high, indicating that non-parallel waveguide distribution can improve positioning accuracy and robustness, which verify our **Remark 5**. In addition, as the number of PAs in the array increases, the average positioning error increases and the lower limit of positioning performance decreases, which verifies our **Remark 4**.

V. Conclusion

In this paper, we first reviewed prior research on PASS and indoor positioning, and then proposed multi-waveguide-based uplink positioning approaches. Two scenarios were considered: MWSP and MWMP scenarios. For the MWSP scenario, an RSSI-based ranging method and an LS positioning algorithm were developed, followed by a detailed error analysis. For the MWMP scenario, we derived a closed-form expression of the superposed signal and designed a grid-search-based positioning algorithm. Furthermore, we conducted the error analysis of the proposed positioning algorithm. Future research may focus on optimizing the signal superposition model to further reduce system errors.

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